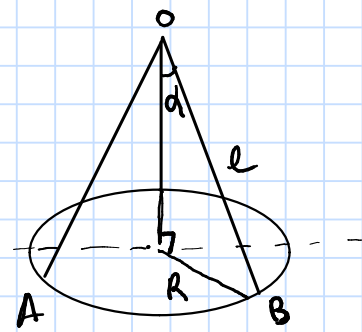
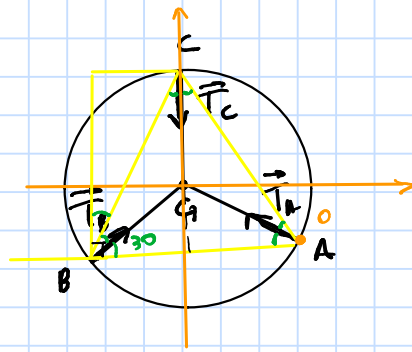
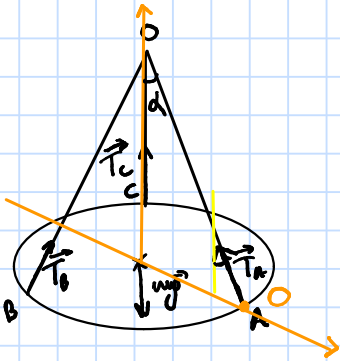




$$\vec{F}^{(a)} = 0 \quad \vec{T}_O^{(e)} = 0$$

$$\vec{T}_A + \vec{T}_B + \vec{T}_C + w\vec{p} = 0$$

$$x) -T_A \cos \frac{\pi}{6} + T_B \cos \frac{\pi}{6} = 0$$

$$T_B = T_A$$

$$y) T_A \sin \frac{\pi}{6} + T_B \sin \frac{\pi}{6} - T_C - w\vec{p} = 0 \quad T_C = (T_A + T_B) \sin \frac{\pi}{6} - w\vec{p}$$

$$\vec{T}_O^{(a)} = 0$$

$$\vec{AG} \times w\vec{p} + \vec{AB} \times \vec{T}_B + \vec{AC} \times \vec{T}_C = 0$$

$$\cancel{R} w\vec{p} \sin \left(\frac{\pi}{2} + \frac{\pi}{6} \right) - \cancel{2R} \cos \frac{\pi}{6} T_B \sin \left(\frac{\pi}{2} + \frac{\pi}{6} \right) + \cancel{2R} \cos \frac{\pi}{6} T_C \sin \left(\frac{\pi}{2} + \frac{\pi}{6} \right) = 0$$

$$R w\vec{p} - 2R \cos \frac{\pi}{6} T_B + 2R \cos \frac{\pi}{6} T_C = 0$$

$$w\vec{p} - 2 \cos \frac{\pi}{6} T_B + 2 \cos \frac{\pi}{6} T_C = 0$$

$$w\vec{p} - T_B + T_C = 0 \rightarrow T_C = T_B + w\vec{p}$$

$$R = \sin \alpha \rightarrow \sin \alpha = \frac{R}{h}$$